MILITARY
SPECIFICATION

MIL-F-14136B(SigC)
18 December 1959

Superseding
MIL-F-14136A(SigC)
30 July 1958

# FILTER, BANDPASS
# (FILTER KITS MK-123( )/TRC, MK-124( )/TRC, MK-228( )/GRC, AND MK-236( )/GRC, UNITS OF)

## 1. SCOPE

1.1 This specification covers requirements for construction of the units of Filter Kits:

MK-123( )/TRC
MK-124( )/TRC
MK-228( )/GRC
MK-236( )/GRC.

These kits contain band pass filters which are used for both Radio Transmitter T-302( )/TRC and Radio Receiver R-417( )/TRC of Intermediate Area Radio Relay Systems. They provide additional selectivity for suppression of noise interference, harmonics, spurious and intermodulation radiations. (See 3.1).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| QQ-S-571 | Solder; Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form. |
| PPP-B-601 | Boxes, Wood, Cleated Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Boxes, Fiber. |
| PPP-T-60 | Tape; Pressure Sensitive Adhesive, Waterproof for Packaging and Sealing. |

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of. |
| MIL-B-117 | Bags, Interior Packaging. |
| MIL-G-3278 | Grease, Aircraft and Instrument (for Low and High Temperature). |
| MIL-M-13231 | Marking of Electronic Items. |
| MIL-F-14072 | Finishes for Ground Signal Equipment. |



FSC 5915

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-169 | Extreme Temperature Cycle. |
| MIL-STD-170 | Moisture Resistance Test Cycle for Ground Signal Equipment. |
| MIL-STD-252 | Wired Equipment, Classification of Visual and Mechanical Defects. |

DRAWINGS

SIGNAL CORPS

| | |
|---|---|
| SC-DL-62698 | Filter, Bandpass F-233( )/U |
| SC-DL-63706 | Filter, Bandpass F-234( )/U |
| SC-DL-63775 | Filter, Bandpass F-235( )/U |
| SC-DL-63784 | Filter, Bandpass F-236( )/U |
| SC-DL-63300 | Filter, Bandpass F-238( )/U |
| SC-DL-63302 | Filter, Bandpass F-239( )/U |
| SC-DL-63307 | Filter, Bandpass F-240( )/U |
| SC-DL-63620 | Filter, Bandpass F-241( )/U |
| SC-DL-63344 | Filter, Bandpass F-242( )/U |
| SC-DL-63320 | Filter, Bandpass F-243( )/U |
| SC-DL-64545 | Case, Accessories CY-1344( )/TRC |
| SC-DL-66525 | Filter, Bandpass F-192( )/U |
| SC-DL-66574 | Filter, Bandpass F-193( )/U |
| SC-DL-66582 | Filter, Bandpass F-194( )/U |
| SC-DL-66590 | Filter, Bandpass F-195( )/U |
| SC-DL-66598 | Filter, Bandpass F-196( )/U |
| SC-DL-66606 | Filter, Bandpass F-197( )/U |
| SC-DL-66614 | Filter, Bandpass F-199( )/U |
| SC-DL-66622 | Filter, Bandpass F-200( )/U |
| SC-DL-66630 | Filter, Bandpass F-201( )/U |
| SC-DL-66638 | Filter, Bandpass F-202( )/U |
| SC-DL-66646 | Filter, Bandpass F-203( )/U |
| SC-DL-66654 | Filter, Bandpass F-204( )/U |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.



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CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification No. 4.

Consolidated Freight Classification Rule No. 21.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Illinois).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification No. A-3 and No. 14.

(Application for copies should be addressed to the National Classification Board 1424 Sixteenth St., NW, Washington 6, D. C.)

U. S. POST OFFICE DEPARTMENT

Parcel Post Regulations

(Application for copies should be obtained from any first class U. S. Post Office).

3. REQUIREMENTS

3.1 Description.- The filters covered by this specification have an input and output impedance of 50 ohms, and are grouped into kits covering operating ranges as follows:

| | |
|---|---|
| 50 to 100 mc | Filter Kit MK-236( )/GRC |
| 100 to 225 mc | Filter Kit MK-123( )/TRC |
| 225 to 400 mc | Filter Kit MK-124( )/TRC |
| 400 to 600 mc | Filter Kit MK-228( )/GRC. |

3.2 Construction.- The items, to be furnished as specified in the bid request and contract, shall be constructed in accordance with the applicable drawings listed below:

MK-236( )/GRC (A Band)

SC-DL-63300 Filter, Bandpass F-238( )/U
SC-DL-63302 Filter, Bandpass F-239( )/U
SC-DL-63307 Filter, Bandpass F-240( )/U
SC-DL-63620 Filter, Bandpass F-241( )/U
SC-DL-63344 Filter, Bandpass F-242( )/U
SC-DL-63320 Filter, Bandpass F-243( )/U
SC-DL-64545 Case, Accessories CY-1344( )/TRC

MK-124( )/TRC (C Band)

SC-DL-66614 Filter, Bandpass F-199( )/U
SC-DL-66622 Filter, Bandpass F-200( )/U
SC-DL-66630 Filter, Bandpass F-201( )/U
SC-DL-66638 Filter, Bandpass F-202( )/U
SC-DL-66646 Filter, Bandpass F-203( )/U
SC-DL-66654 Filter, Bandpass F-204( )/U
SC-DL-64545 Case, Accessories CY-1344( )/TRC



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MK-123( )/TRC (B Band)

| | |
|---|---|
| SC-DL-66525 | Filter, Bandpass F-192( )/U |
| SC-DL-66574 | Filter, Bandpass F-193( )/U |
| SC-DL-66582 | Filter, Bandpass F-194( )/U |
| SC-DL-66590 | Filter, Bandpass F-195( )/U |
| SC-DL-66598 | Filter, Bandpass F-196( )/U |
| SC-DL-66606 | Filter, Bandpass F-197( )/U |
| SC-DL-64545 | Case, Accessories CY-1344( )/TRC |

MK-228( )/GRC (D Band)

| | |
|---|---|
| SC-DL-62698 | Filter, Bandpass F-233( )/U |
| SC-DL-63706 | Filter, Bandpass F-234( )/U |
| SC-DL-63775 | Filter, Bandpass F-235( )/U |
| SC-DL-63784 | Filter, Bandpass F-236( )/U |
| SC-DL-64545 | Case, Accessories CY-1344( )/TRC. |

3.3 Lubricant.- A lubricant, in accordance with Specification MIL-G-3278, shall be used to lubricate the dials and the tuning shafts of all filters at the front panel bushing.

3.4 Finish protective.- Units shall be given a protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware such as handles, hinges, screws, etc., and necessary touch up after mounting.

3.5 Marking of items.- Marking shall conform to Specification MIL-M-13231. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with a minimum disassembly of the items in which they are used. Front panel marking and dial calibrations of the filters shall be Group I, pigment filled.

3.5.1 Serial numbers.- Serial numbers under Specification MIL-M-13231 are not required.

3.6 Soldering.-

3.6.1 Solder.- Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition SN60, type AR or S, shall be used for general purposes. When type S solder is used for soldering electrical connections, a rosin type flux shall be employed and shall meet the applicable requirements of Specification QQ-S-571 for the core of type AR solder.

3.6.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for, or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.6.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the unit; flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.



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3.7 Cleaning.-

3.7.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.7.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.8 Interchangeability.- The contractor's procedures and methods--including engineering, production, and quality control--shall insure interchangeability of parts as specified below.

3.8.1 Corresponding parts.- "Corresponding parts" as used herein, in connection with the interchangeability requirements, means all possible parts meeting a set of limiting values (dimensions, tolerances, ratings and limits). Such values shall be those specified in the applicable Government documents, supplemented as necessary with values established by the manufacturer of the part; however, if no Government document applies, the limiting values shall be those established by the manufacturer of the part.

3.8.2 Interchangeability requirement.- Any part shall be replaceable with any corresponding part without degrading electrical, mechanical, or physical characteristics and performance below specified values and without modification of the replacement part is used. It is permissible, however, to use normal adjustments in meeting this requirement. (The foregoing requirement necessarily precludes reliance on selective assembly to meet specified requirements, or reliance on any characteristic not included in the limiting values of the part).

3.9 Electrical requirements.- (See 4.9).

3.9.1 Dial calibration accuracy.- The dial marking calibration shall be accurate to within ± 3 channels of the tuned channel for all channels within the range of each filter. (The dial markings are expressed both in channel numbers and frequencies).

3.9.2 Insertion loss.- The attenuation shall be in accordance with the following tables:

Table Ia

Filter Kit MK-236( )/TRC

| Bandpass Filter | Useful Frequency Range (MC) | Test Frequency F(MC) | Maximum Loss At F(db) | Loss At F-8% F mc (db) | Loss At F+8% F mc (db) |
|---|---|---|---|---|---|
| F-238( )/U | 50-58.5 | 54.125 | 1.0 | 24. ± 3 | 24. ± 3 |
| F-239( )/U | 58.5-67 | 62.625 | 1.0 | 26. ± 3 | 27. ± 3 |
| F-240( )/U | 66.8-75.5 | 71.125 | 1.0 | 28. ± 3 | 30. ± 3 |
| F-241( )/U | 75-84 | 79.375 | 1.0 | 30. ± 3 | 32. ± 3 |
| F-242( )/U | 83.5-92.5 | 88.125 | 1.0 | 32. ± 3 | 35. ± 3 |
| F-243( )/U | 92.6-100 | 96.625 | 1.0 | 33. ± 3 | 37. ± 3 |



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Table Ib

Filter Kit MK-123( )/TRC

| Bandpass Filter | Useful Frequency Range (MC) | Test Frequency F(MC) | Maximum Loss at F(db) | Loss At F-15 mc (db) | Loss At F+12 mc (db) |
|---|---|---|---|---|---|
| F-192( )/U | 100-121 | 110 | 1.0 | 36. ± 3 | 37. ± 4 |
| F-193( )/U | 121-142 | 131 | 1.0 | 36. ± 3 | 38. ± 4 |
| F-194( )/U | 142-163 | 152 | 1.0 | 37. ± 3 | 39. ± 4 |
| F-195( )/U | 163-184 | 173 | 1.0 | 37. ± 3 | 39. ± 4 |
| F-196( )/U | 184-205 | 194 | 1.0 | 37. ± 3 | 39. ± 4 |
| F-197( )/U | 205-226 | 215 | 1.0 | 37. ± 3 | 38. ± 4 |

Table Ic

Filter Kit MK-124( )/TRC

| Bandpass Filter | Useful Frequency Range (MC) | Test Frequency F(MC) | Maximum Loss At F(db) | Loss At F-20 mc (db) | Loss At F+15 mc (db) |
|---|---|---|---|---|---|
| F-199( )/U | 223-253 | 238 | 1.0 | 36. ± 3 | 37. ± 4 |
| F-200( )/U | 253-283 | 268 | 1.0 | 36. ± 3 | 36. ± 4 |
| F-201( )/U | 283-313 | 298 | 1.0 | 36. ± 3 | 35. ± 4 |
| F-202( )/U | 313-343 | 328 | 1.0 | 36. ± 3 | 35. ± 4 |
| F-203( )/U | 343-373 | 358 | 1.0 | 36. ± 3 | 34. ± 4 |
| F-204( )/U | 373-403 | 388 | 1.0 | 35. ± 3 | 33. ± 4 |

Table Id

Filter Kit MK-228( )/GRC

| Bandpass Filter | Useful Frequency Range (MC) | Test Frequency F(MC) | Maximum Loss At F(db) | Loss At F -5%F mc (db) | Loss At F+5%F mc (db) |
|---|---|---|---|---|---|
| F-233( )/U | 400-450 | 425 | 1.0 | 31. ± 4 | 31. ± 4 |
| F-234( )/U | 450-500 | 475 | 1.0 | 31. ± 4 | 32. ± 4 |
| F-235( )/U | 500-550 | 525 | 1.0 | 32. ± 4 | 33. ± 4 |
| F-236( )/U | 550-600 | 575 | 1.0 | 33. ± 4 | 34. ± 4 |

3.9.3 Frequency overlap.- Each of the above filters shall have an additional frequency overlap of 3 r-f channels beyond the nominal frequency range at both the low and high end of each filter with a maximum insertion loss of 1.0 db.



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3.10 Service conditions.- The items covered by this specification shall meet the following service conditions:

3.10.1 Temperature (see 4.10).-

(a) Operating: Ambient temperature in the range of + 150°F to -40°F. Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.10.2 Relative humidity (see 4.11).- Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.10.3 Immersion; Case, Accessory CY-1344( )/TRC, (see 4.12).- Three feet of water for two hours.

3.10.4 Bounce and shock.- As described in 3.11.

3.11 Bounce and shock.-

3.11.1 Bounce.- After subjection to the test of 4.13, the filter kits shall meet full specification requirements. Minor surface abrasion is allowed.

3.11.2 Shock; drop.- After subjection to the test of 4.14, the filter kits shall meet full specification requirements, with only minor mechanical damage permissable.

3.12 Preconditioning. The unit shall be capable of meeting the inspection of Section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.5. (Also see 4.4(d)).

3.13 Preproduction samples.- The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract. (See 6.2(e)(1)).

3.14 Technical literature.- Technical literature shall be furnished as specified in the contract.

3.15 Workmanship.- The units on order shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.2 Construction
3.3 Lubricant
3.4 Finish, protective
3.5 Marking of items
3.6 Soldering
3.7 Cleaning



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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3).

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.4 and 4.15).

(1) Procurement inspection of items before preparation for delivery. (See 4.4).

(2) Procurement inspection of preparation for delivery. (See 4.15 and Section 5).

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table II and the Group A, Group B, and Group C inspections specified in Tables III, IV, and V, respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) bounce, (2) shock, and (3) immersion; other preproduction inspection may preceed, follow, or be interspersed between the foregoing.

Table II - Preproduction inspection

| Inspection (for additional preproduction inspection see 4.3) | Requirement Paragraph | Inspection Paragraph |
|---|---|---|
| Temperature | 3.10.1 | 4.10 |
| Moisture-resistance | 3.10.2 | 4.11 |
| Bounce | 3.11.1 | 4.13 |
| Shock; drop | 3.11.2 | 4.14 |

4.4 Procurement inspection of items before preparation for delivery.- The contractor shall perform the inspection specified by (a) through (d) below, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition



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4.4 Procurement inspection of items before preparation for delivery.- (Contd)

the Government--at its discretion--may perform all or any part of the inspection specified in (a), to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the item on contract meets the specified procurement inspection. (Also see 6.6).

(a) Procurement inspection shall consist of group A, group B, and group C inspection as specified in 4.4.1 through 4.4.3.2.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when it is considered more practical to select separate samples for group B or group C inspection, or both.

(d) Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly. (See 4.5).

4.4.1 Group A inspection.- This inspection (including sampling) shall conform to table III and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group A inspection

| Inspection | Rqt par. | Insp. par. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Visual and mechanical | | | | |
| Filters | 3.15 | 4.6 | 2.5 dphu* | 6.5 dphu |
| Complete Kits, (incl. cases) | 3.15 | 4.6 | 2.5 dphu | 6.5 dphu |
| Interchangeability | 3.8 | 4.7 | 1.0% | |
| Electrical | | | | |
| Dial calibration | 3.9.1 | 4.9.1 | 1.0% (combined) | ** |
| Insertion loss | 3.9.2 | 4.9.2 | | |

* dphu - Defects per hundred units.
** All electrical defects are considered major.



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4.4.2 Group B inspection.- This inspection (including sampling) shall conform to Table IV and MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedures shall be R-1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.4.5 and the requirements of Standard MIL-STD-105 for disposition of rejected product. The AQL shall be 10 percent defective and the inspection level shall be L-6 for normal and tightened inspection and L-4 for reduced inspection.

4.4.2.1 Procedure in case of failure.- When an inspection lot fails, the contractor shall immediately investigate the cause of failure and take corrective action to assure that subsequent lots do not contain the same defect or defects.

Table IV - Group B Inspection

| Inspection | Rqt Par. | Insp Par |
|---|---|---|
| Immersion (Case CY-1344( )/TRC) | 3.10.3 | 4.12 |

4.4.3 Group C inspection.- This inspection shall be as listed in Table V.

4.4.3.1 Sampling for inspection of unit.- One filter of each type produced shall be subjected each month to the group C inspection. This test shall be performed on sample filters produced during that month. (See 4.4(c)).

4.4.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.3).

Table V - Group C inspection

| Inspection | Requirement Paragraph | Inspection Paragraph |
|---|---|---|
| Frequency overlap | 3.9.3 | 4.9.3 |



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4.4.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.4.5 Disposition of nonconforming product.- The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A, B, or C inspection.

Inspection lots which failed during group A or B inspection.

Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 Bounce preconditioning.- Each individual filter shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the filter to a distance of 3 inches as the unit bounces. The filter shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the filter shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.6 Visual and mechanical inspection.- The items on order shall be examined for defects listed in Standard MIL-STD-252.

4.7 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

Filters.-

(a) Location of rails, guide-pin holes, and connectors.

(b) Outer dimensions of filter unit.

Case CY-1344( )/TRC.-

(a) Inside and outside dimensions of the transit case and cover.

(b) Outside dimensions of integral case.

(c) Dimensions of filter cavities in integral case.

(d) Location of catches on case and cover.



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4.8 Test circuit requirements.- The test set used to test the performance of the filters shall meet the following requirements:

(a) Frequency accuracy: ± 0.1 mc for frequencies 40 to 650 mc inclusive.

(b) Insertion loss accuracy:

| Range DB | A Band | B Band | C Band | D Band |
|---|---|---|---|---|
| 0 to 5 | ± 0.1 db | ± 0.1 db | ± 0.1 db | ± 0.2 db |
| 5 to 10 | ± 0.1 db | ± 0.1 db | ± 0.2 db | ± 0.3 db |
| 10 to 20 | ± 0.3 db | ± 0.3 db | ± 0.4 db | ± 0.6 db |
| 20 to 30 | ± 0.6 db | ± 0.7 db | ± 0.8 db | ± 1.2 db |
| 30 to 40 | ± 0.6 db | ± 0.7 db | ± 0.8 db | ± 1.2 db |

(c) Generator impedance: (50 ± 1) + j (0 ± 1) ohms.

(d) Detector impedance: (50 ± 1) + j (0 ± 1) ohms.

(e) The combined discrimination of the generator and receiver against harmonics of the generator including its oscillator shall be at least 60 db.

(f) Differential sensitivity. The metering circuit shall respond to changes of at least 1/2 the accuracy for insertion loss.

4.9 Inspection tests.- The test equipment shall be connected as shown on Figure 1, and the filters shall be tested under standard test conditions as follows:

4.9.1 Dial Calibration.-

4.9.1.1 Using the frequency meter, adjust the frequency of the signal generator to test frequency (a) Table VI.

4.9.1.2 Switch to the filter under Test and adjust the filter dials to a maximum indication on the meter.

4.9.1.3 The dials on the filter shall be within ± 3 channels of the channel listed in Table VI for test frequency (a).

4.9.1.4 Repeat steps 4.9.1.1 and 4.9.1.2 above for test frequencies (b & c) Table VI.

4.9.1.5 The dials on the filter shall be within ± 3 channels of the channel listed in Table VI for test frequencies (b & c).

4.9.2 Insertion loss.-

4.9.2.1 Using the frequency meter, adjust the frequency of the signal generator to test frequency (d) Table VI for the filter under test.



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4.9.2.2 Switch to the filter under test, and adjust its dials for a maximum deflection on the meter. Note the reading on the meter. Do not readjust filter during remainder of test.

4.9.2.3 Switch to the step attenuator, and adjust the attenuator for the same reading as noted in step 4.9.2.2 above.

4.9.2.4 The insertion loss of the filter shall be less than indicated in Table I for test frequency (d) of Table VI.

4.9.2.5 Switch to the filter under test.

4.9.2.6 Adjust the frequency of the signal generator to test frequency (e) Table VI.

4.9.2.7 Note the reading on the meter.

4.9.2.8 Switch to the step attenuator position adjust attenuator to th same reading as in step 4.9.2.7 above.

4.9.2.9 The insertion loss shall be as indicated in Table I for Test frequency (e) of Table VI.

4.9.2.10 Repeat 4.9.2.5 through 4.9.2.8 for test frequency (f) Table VI.

4.9.2.11 The insertion loss shall be as indicated in Table I for test frequency (f) of Table VI.

4.9.3 Frequency overlap.- Repeat 4.9.2 above except that the test frequencies shall be: Table VI, (a) -3 channels and (b) +3 channels. (A Band: 1 channel = .250 mc, B Band: 1 channel = .50 mc., C Band: 1 channel = 1.0 mc., and D Band: 1 channel = 1.50 mc).



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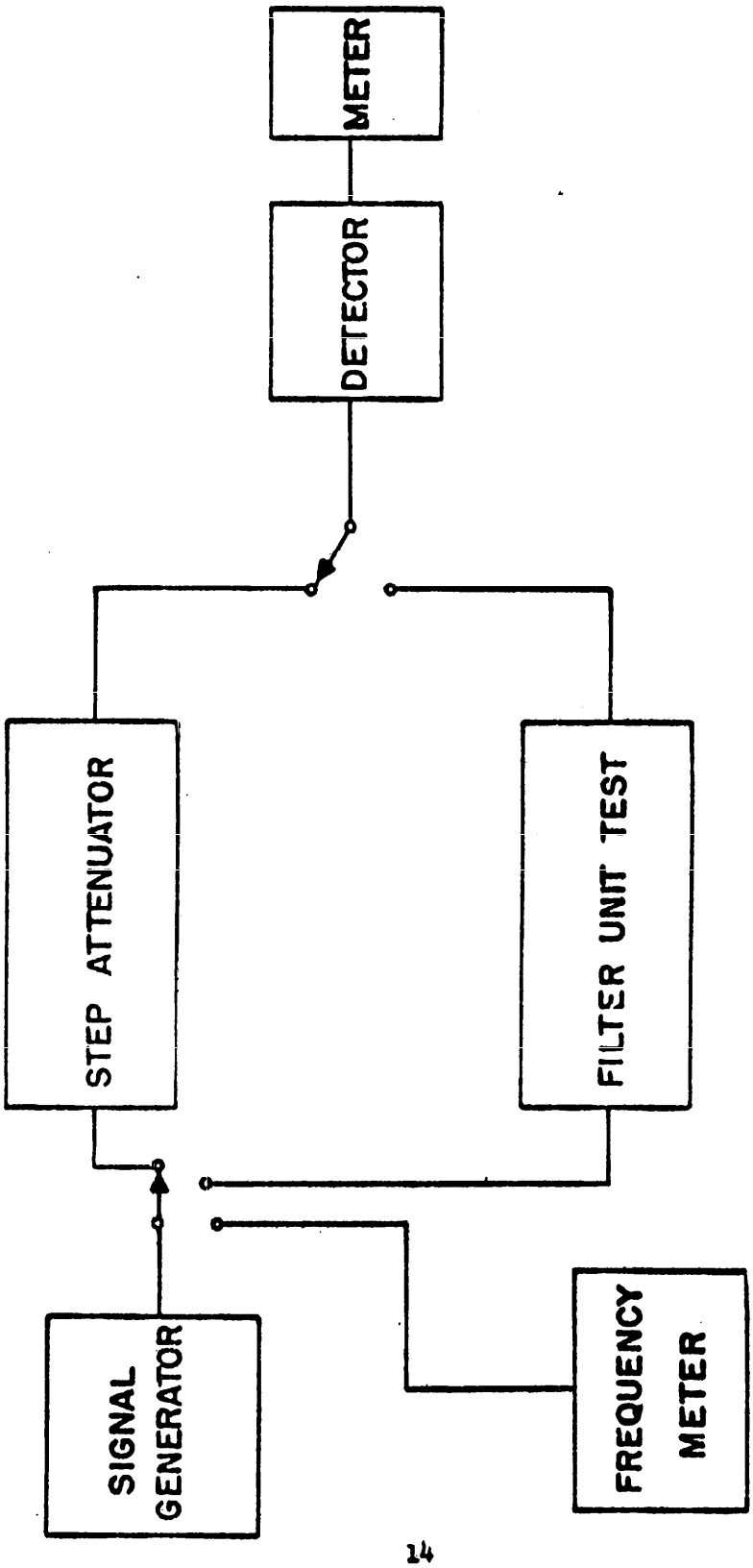



FIG. 1



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Table VI - Test frequencies

| Bandpass Filter | Channel No. | Test Frequency MC |
|---|---|---|
| A Band: | | |
| F-238( )/U | (a) 1 | 50.125 |
| | (b) 34 | 58.375 |
| | (c) 17 | 54.125 |
| | (d) | 54.125 |
| | (e) | 49.795 |
| | (f) | 58.455 |
| F-239( )/U | (a) 34 | 58.375 |
| | (b) 68 | 66.875 |
| | (c) 51 | 62.625 |
| | (d) | 62.625 |
| | (e) | 57.615 |
| | (f) | 67.635 |
| F-240( )/U | (a) 68 | 66.875 |
| | (b) 102 | 75.375 |
| | (c) 85 | 71.125 |
| | (d) | 71.125 |
| | (e) | 64.435 |
| | (f) | 76.815 |
| F-241( )/U | (a) 102 | 75.375 |
| | (b) 136 | 83.875 |
| | (c) 118 | 79.375 |
| | (d) | 79.375 |
| | (e) | 73.025 |
| | (f) | 85.725 |
| F-242( )/U | (a) 136 | 83.875 |
| | (b) 170 | 92.375 |
| | (c) 153 | 88.125 |
| | (d) | 88.125 |
| | (e) | 81.075 |
| | (f) | 95.175 |
| F-243( )/U | (a) 170 | 92.375 |
| | (b) 201 | 100.125 |
| | (c) 187 | 96.625 |
| | (d) | 96.625 |
| | (e) | 88.895 |
| | (f) | 104.355 |



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Table VI - Test frequencies (Contd)

| Bandpass Filter | Channel No. | Test Frequency MC |
|---|---|---|
| B Band: | | |
| F-192( )/U | (a) 2 | 100.75 |
| | (b) 40 | 119.75 |
| | (c) 20 | 109.75 |
| | (d) | 110.00 |
| | (e) | 95.00 |
| | (f) | 122.00 |
| F-193( )/U | (a) 44 | 121.75 |
| | (b) 82 | 140.75 |
| | (c) 62 | 130.75 |
| | (d) | 131.00 |
| | (e) | 116.00 |
| | (f) | 143.00 |
| F-194( )/U | (a) 86 | 142.75 |
| | (b) 124 | 161.75 |
| | (c) 120 | 150.75 |
| | (d) | 152.00 |
| | (e) | 137.00 |
| | (f) | 164.00 |
| F-195( )/U | (a) 128 | 163.75 |
| | (b) 166 | 182.75 |
| | (c) 144 | 171.75 |
| | (d) | 173.00 |
| | (e) | 158.00 |
| | (f) | 185.00 |
| F-196( )/U | (a) 170 | 184.75 |
| | (b) 208 | 203.75 |
| | (c) 184 | 191.75 |
| | (d) | 194.00 |
| | (e) | 179.00 |
| | (f) | 206.00 |
| F-197( )/U | (a) 212 | 205.75 |
| | (b) 250 | 224.75 |
| | (c) 226 | 212.75 |
| | (d) | 215.00 |
| | (e) | 200.00 |
| | (f) | 227.00 |



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Table VI - Test frequencies (Contd)

| Bandpass Filter | Channel No. | Test Frequency MC |
|---|---|---|
| C Band: | | |
| F-199( )/U | (a) 26 | 225.50 |
| | (b) 52 | 251.50 |
| | (c) 32 | 231.50 |
| | (d) | 238.00 |
| | (e) | 218.00 |
| | (f) | 253.00 |
| F-200( )/U | (a) 56 | 255.50 |
| | (b) 82 | 281.50 |
| | (c) 62 | 261.50 |
| | (d) | 268.00 |
| | (e) | 248.00 |
| | (f) | 283.00 |
| F-201( )/U | (a) 86 | 285.50 |
| | (b) 112 | 311.50 |
| | (c) 92 | 291.50 |
| | (d) | 298.00 |
| | (e) | 278.00 |
| | (f) | 313.00 |
| F-202( )/U | (a) 116 | 315.50 |
| | (b) 142 | 341.50 |
| | (c) 122 | 321.50 |
| | (d) | 328.00 |
| | (e) | 308.00 |
| | (f) | 343.00 |
| F-203( )/U | (a) 146 | 345.50 |
| | (b) 172 | 371.50 |
| | (c) 152 | 351.50 |
| | (d) | 358.00 |
| | (e) | 338.00 |
| | (f) | 373.00 |
| F-204( )/U | (a) 176 | 375.50 |
| | (b) 202 | 401.50 |
| | (c) 182 | 381.50 |
| | (d) | 388.00 |
| | (e) | 368.00 |
| | (f) | 403.00 |



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Table VI - Test frequencies (Contd)

| Bandpass Filter | Channel No. | Test Frequency MC |
|---|---|---|
| D Band: | | |
| F-233( )/U | (a) 67 | 399.750 |
| | (b) 101 | 450.750 |
| | (c) 84 | 425.250 |
| | (d) | 425.250 |
| | (e) | 403.9875 |
| | (f) | 446.5725 |
| F-234( )/U | (a) 101 | 450.750 |
| | (b) 134 | 500.250 |
| | (c) 118 | 476.250 |
| | (d) | 476.250 |
| | (e) | 452.4375 |
| | (f) | 500.0625 |
| F-235( )/U | (a) 134 | 500.250 |
| | (b) 167 | 549.750 |
| | (c) 151 | 525.750 |
| | (d) | 525.750 |
| | (e) | 499.4625 |
| | (f) | 552.0375 |
| F-236( )/U | (a) 167 | 549.750 |
| | (b) 200 | 599.250 |
| | (c) 184 | 575.250 |
| | (d) | 575.250 |
| | (e) | 546.4875 |
| | (f) | 604.0125 |

4.10 Temperature.- The filters shall be subjected to the temperature cycle shown on Standard MIL-STD-169 and the tests of 4.9 performed at steps 3 and 8.

4.11 Moisture-resistance test.

4.11.1 Test conditions.-

(a) Do not move the filters from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.11.2 Test method.- The filters shall be tested as follows:



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4.11.2 Test method.- (Contd)

(a) Dry at 130° ± 5°F for 24 hours.

(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.9 and readjust or realign as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. The measurements specified in 4.9 shall be made during the periods shown on the standard and shall comply with 4.9.

(e) After cycling has been completed, condition the filters for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the filters. No repair of replacement or parts shall be made. After adjustment, the filters shall meet full specified performance for those measurements specified in 4.9.

4.12 Immersion test.- Case CY-1344( )/TRC, without its inner shock frame, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the case shall be 40°F, or more, above the temperature of the water. The tank in which the case is immersed shall be of sufficient capacity to maintain the water within ±2°F of its initial temperature, or the temperature of the water shall be maintained within these limits by other means. After completion of the 2-hour period of immersion, the case shall be removed from the water and wiped dry on exterior surfaces. When the case is opened, there shall be no evidence of leakage.

4.13 Bounce test.- The filters, in Case CY-1344( )/TRC, shall be tested on the package tester, as made by the L.A.B. Corporation, Skaneateles, New York, or equal, as follows:

(a) Cover the tester bed with a panel of 1/2 inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6 inch intervals around all four edges and at 3 inch intervals in a 6 inch square in the center.

(b) Place the case on the bed of the package tester, limiting the lateral motion (by wooden fences) to nor more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the case against the back stop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ± 2 rpm. Turn the case at the end of each 30 minutes so it will rest on a new face.

4.14 Drop Test.- The filters, in Case CY-1344( )/TRC, shall be dropped a distance of two feet on each corner, edge and face (total of 26 drops). Drops shall be made from a quick-release hook; an Acme Drop Tester, or equal; or a basket-type pendulum tester. The floor or barrier receiving the impact shall be a 2-inch fir, backed by concrete or a rigid steel frame.



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4.15 Procurement inspection of preparation for delivery.- Inspection of preparation for delivery shall conform to Specification MIL-P-116.

4.16 Rough handling test (preparation for delivery).- When rough handling test, in accordance with Specification MIL-P-116, is required by the contract, the tests of 4.9.1 shall be conducted to determine freedom from operational malfunction caused by the rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Filter, Bandpass, (Filter Kits MK-123( )/TRC, MK-124( )/TRC, MK-228( )/GRC and MK-236( )/GRC) shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Technical manual.- Each technical manual shall be packaged Method IC-3 as follows: Inclose each technical manual within a close-fitting Type II, Class b bag conforming to Specification MIL-B-117. Heat seal the bag to effect closures.

5.1.1.2 Filter Bandpass in Case CY-1344( )/U.-

5.1.1.2.1 Unit package.- Each filter bandpass in case shall be unit packaged Method IC-5 as follows: Cushion each case on all surfaces with cells or pads or both fabricated of double faced corrugated fiberboard, having a minimum dry bursting strength of 200 pounds and designed to absorb the shock of impact encountered in handling and transit. Cells or pads shall have water-resistant qualities equal to the box in which they shall be used. Place the cushioned case with contents within a close-fitting regular slotted style Class 2 W5c fiber box conforming to Specification PPP-B-636, Table II. Box closure shall be in accordance with the appendix of the referenced specification. In addition seal all corners, seams and joints with Type III, Class 1 olive drab pressure sensitive waterproof tape conforming to Specification PPP-T-60.

5.1.2 Level C.- Filter Bandpass (Filter Kits MK-123( )/TRC, MK-124( )/TRC, MK-228( )/GRC, and MK-236( )/GRC) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- Filter Bandpass (Filter Kits MK-123( )/TRC, MK-124( )/TRC, MK-228( )/GRC and MK-236( )/GRC) packaged as specified in 5.1.1.2 shall be packed for shipment one each within a style A, Type I wood, cleated plywood box conforming to Specification PPP-B-601, or box, wood, nailed and lock-corner, Class 2, Style 2 conforming to Specification PPP-B-621. Fabricate the box to fit the contents snugly. Place the technical manual (see 5.1.1.1) between the contents and lid of the box. Box closure shall be in accordance with the appendix of the applicable box specification.



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5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.2.2 Level B.- Filter, Bandpass (Filter Kits MK-123( )/TRC, MK-124( )/TRC, MK-228( )/GRC and MK-236( )/GRC) shall be packed for shipment as specified in 5.2.1 except boxes shall be domestic type or style.

5.2.3 Level C.- Filter, Bandpass (Filter Kits MK-123( )/TRC, MK-124( )/TRC, MK-228( )/GRC, and MK-236( )/GRC) shall be packed for shipment in a manner conforming to Uniform Freight Classification No. 4 and Consolidated Freight Classification, Rule 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.13).

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon.

(f) Marking and shipping of samples.



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(g) Place of final inspection.

(h) Technical literature required. (See 3.14).

6.2 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: MK-123W/TRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.3 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.4.3.2).

6.4 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.4.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.5 Equipment divisions.- The terms used herein for equipment divisions conform to Standard MIL-STD-280.

6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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